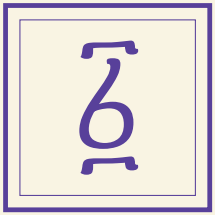
# Computing clusters

*The cost of computation and supercomputers*

Carlos Cotrini
*AI in Industry, ETH Zurich*
*Saturday 14 November 2026* 𓂀𓆙𓂀𓆙

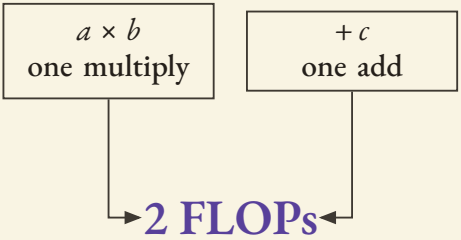


*Section 1*

# One operation, and who does it

# A FLOP is one operation on two numbers

**One multiply accumulate**



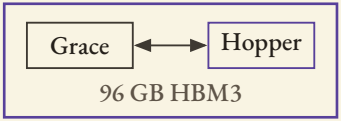

**Count, or rate**

**FLOP:** a count of operations
*how much work*

**FLOP/s:** operations per second
*how fast*

# One computer, then and now



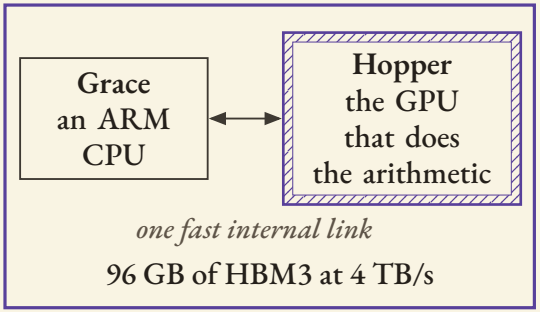



| | Human computer, 1949 | One GH200, 2024 |
|---|---|---|
| *Rate* | 1,000 additions a day | $990 \times 10^{12}$ FLOP/s |
| *Format* | Decimal, by hand | 16 bits per number |
| *Power* | About 100 W | 660 W on Alps |
| *Food* | **2,065 kcal a day**<br>*what one person eats* | **13,600 kcal a day**<br>*what six or seven people eat* |

The clerk's rate is what the shop *delivered*. The chip's is what it *could* do.

*Photograph: a computer room, NACA, 1949 (NASA Dryden, E49-54), public domain.*

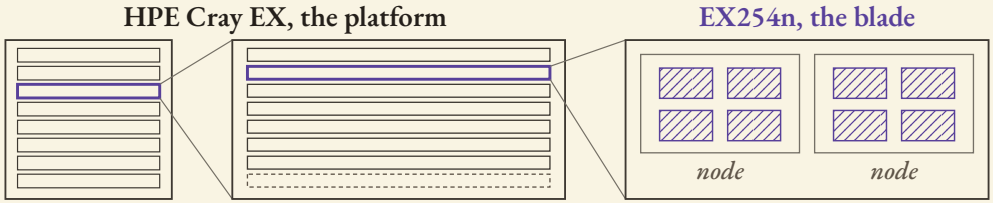
# A GH200 is a CPU and a GPU on one module




- *The GPU computes. The CPU reads the data, runs Python, and keeps it fed*
- *HBM3 is memory stacked on the GPU package: only 96 GB, but 4 TB/s*
- *Between them 900 GB/s, about seven times what a plug-in card gets*

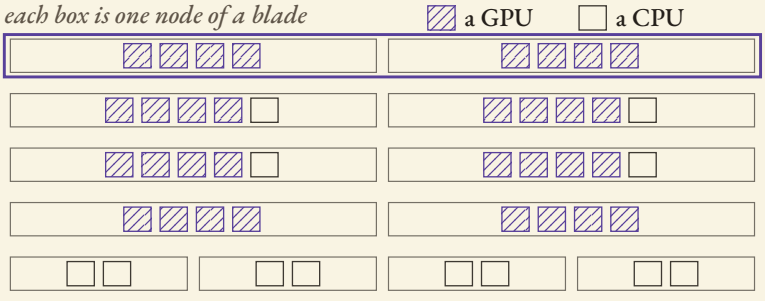
# Alps is an HPE Cray EX254n

The TOP500 entry reads: *Alps, HPE Cray EX254n, NVIDIA Grace 72C 3.1GHz, NVIDIA GH200 Superchip, Slingshot-11, HPE Cray OS*




The platform: cabinets, liquid cooling, power and the network.

# One platform, several blade types



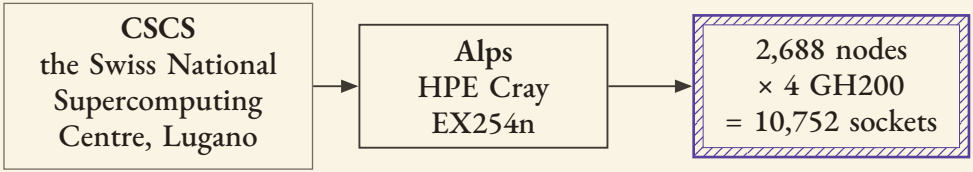

*Every name here that ends in n carries NVIDIA GPUs.*
The blade choice is what makes Alps a machine for AI rather than a conventional CPU cluster.

# The atelier against Alps

| A collection of computers | Rate | Power |
|---|---|---|
| Prony's atelier, 1795<br>twenty people | 0.694 additions/s | about 2 kW |
| Alps, 2024<br>10,752 modules | $10.64 \times 10^{18}$ FLOP/s | 7.1 MW |

**Fifteen billion billion times the rate,**
**on three and a half thousand times the power**

# CSCS and Alps, the machine behind Apertus




**10.6 EFLOP/s** $= 10.6 \times 10^{18}$ **FLOP/s**

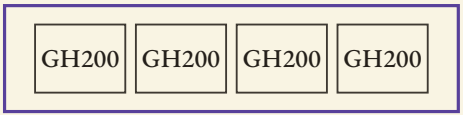
*all 10,752 GH200 together*

Apertus, the open language model of EPFL, ETH Zurich and CSCS, was trained on Alps.

*Machine: cscs.ch/computers/alps and top500.org/system/180259. Model: arXiv 2509.14233.*

# Nodes and sockets

One node




*One operating system, one address space*

**Socket** one processor module

**Node** one independent computer

2,688 nodes × 4 modules = 10,752 sockets

Socket to socket, inside the node: **150 GB/s**

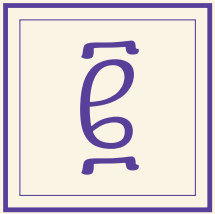
Between nodes: **25 GB/s per GPU, six times less**

The node boundary is where the bandwidth falls off.

# One second of a GH200, in three registers

| The same one second of arithmetic | Time |
|---|---|
| The mean age of all 8.2 billion people alive, one addition each | 8.3 microseconds |
| 130,000 photographs classified by ResNet-50, each 224 by 224 pixels, or 0.05 megapixels | one second |
| Prony's atelier doing that same one second of work | 45 to 136 million years |

*ResNet-50: He and others, arXiv 1512.03385: 3.8 billion multiply-adds, so* $7.6 \times 10^9$ *FLOP per image. A peak count.*



*Section 2*

# What the headline rate hides

# Capacity and bandwidth are different numbers

**96 GB**
how much fits

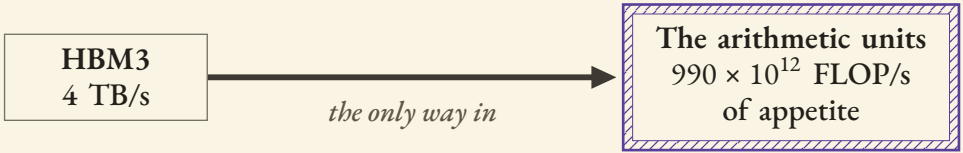
Decides whether the weights and the optimiser state fit on one device at all

**4 TB/s**
how fast you can move it

Decides how quickly the arithmetic units can be fed once the data is there

96 GB is less than an ordinary server holds. That is the trade: HBM is stacked on the package and wired short and wide, which is what buys the speed.

# Two hundred and forty six operations per byte

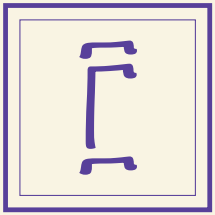



$$\frac{990 \times 10^{12}\ \text{FLOP/s}}{4.023 \times 10^{12}\ \text{bytes/s}} = 246\ \text{FLOP per byte}$$

*A rate over a rate: the seconds cancel, and what is left is operations per byte*

So unless the computation reuses each byte about that many times, the units wait. A large matrix multiply does reuse. Many other operations do not.

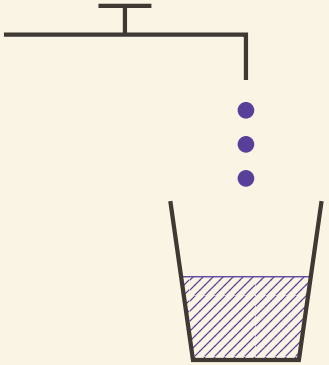
This is one of the reasons a real run reaches 21 to 46 percent of peak.



*Section 3*

# The machine, and what it draws

# A watt is a rate, a kilowatt hour is an amount



**660 W**

How fast energy arrives

**0.66 kWh** *after one hour*

How much arrived

# Watch the slash

**km/h** The slash divides, so it is a speed.

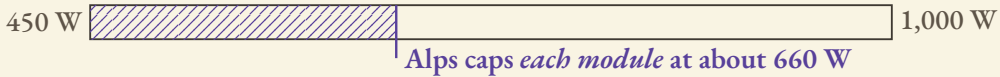
**kWh** No slash. The hour **multiplies**, so it is an amount.

Longer gives you *more*, so the hour cannot be dividing.

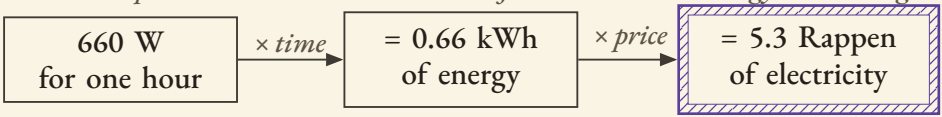
| rate | × | time | = | amount |
|---|---|---|---|---|
| FLOP/s | × | seconds | = | FLOP |
| watts | × | hours | = | watt hours |

# A wattage becomes a bill in two multiplications

**One module is just a band**



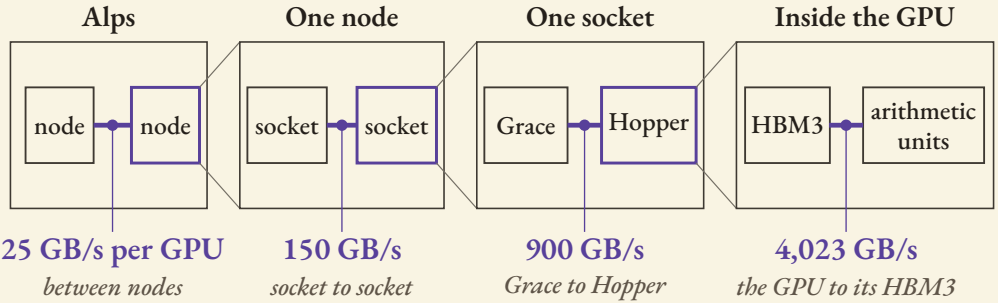

*The operator chooses: near 1,000 W is faster, lower saves energy and cooling*




That is one module. All 10,752 is 7.1 megawatts, about 570 francs an hour.
The whole Apertus run is 3.96 GWh, about 0.3 million francs.

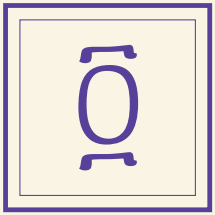
*450 to 1,000 W: GH200 datasheet. 660 W: the Alps cap, docs.cscs.ch/running/slurm. 8 Rappen per kWh assumed (ElCom 2026, large consumers 6.75 to 9; households pay 27.7). Check: 10,752 × 660 W is 7,096 kW, against the 7,124 kW the TOP500 measured.*

# Four links, and each step down costs you




**Each step down costs about a factor of five. End to end, 161 times.**

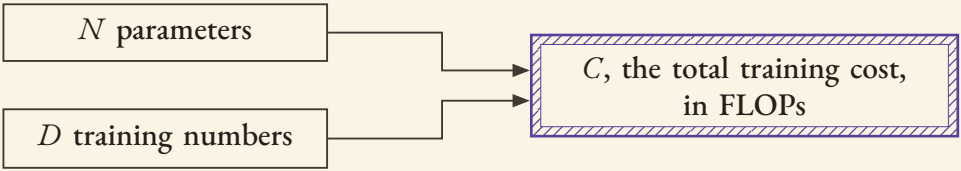
So you keep the data at the fast end and cross the slow links as rarely as you can.



*Section 4*

# From a model to GPU hours

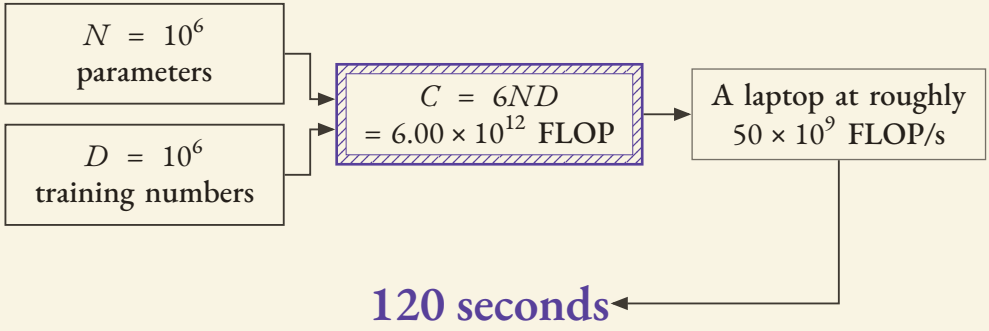
# Two numbers price a training run



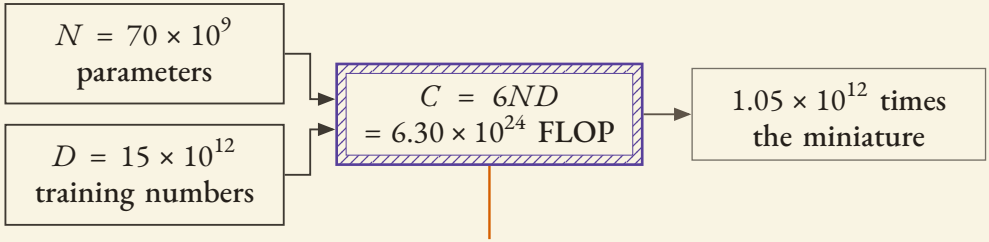

$$C = \underbrace{6}_{\text{per parameter, per training number}} \times \underbrace{N}_{\text{parameters}} \times \underbrace{D}_{\text{training numbers}}$$

*Kaplan and others, 2020, section 2.1. Here the 6 is a rule of thumb.*

# A million by a million is two minutes



# The same formula on Apertus-70B




*Model: arXiv 2509.14233.*

# From one GPU hour to the whole run

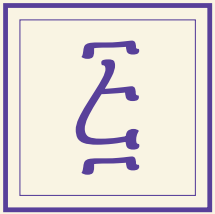
The run: training Apertus-70B on 15 trillion training numbers took over 6 million GPU hours, on up to 4,096 GPUs.

| 4,096 GPUs flat out for **61 days** | It spanned about 91 days, so **2,747** on average |
|---|---|

*The gap is ramp up, restarts and smaller phases: a campaign rarely runs at full width without a break*

6 million GPU hours × 660 W = **3.96 GWh**
× 8 Rappen = **0.32 million francs**

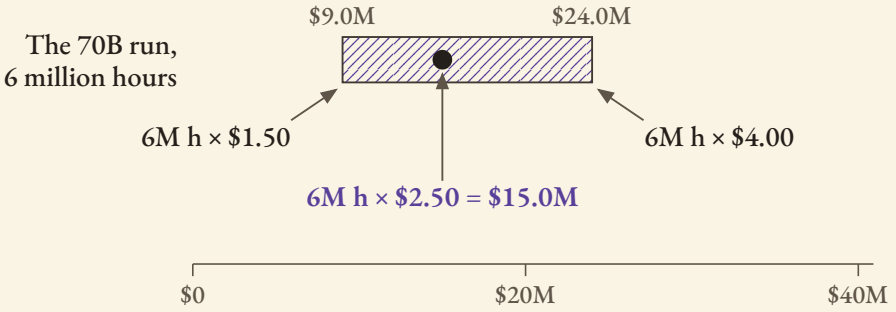
*Run: arXiv 2604.12973.*



*Section 5*

# Dollars and energy

# Six million hours, priced at cloud rates




*The $1.50 to $4.00 band is an H100 rental survey, November 2025.*

# The same hours, in gigawatt hours

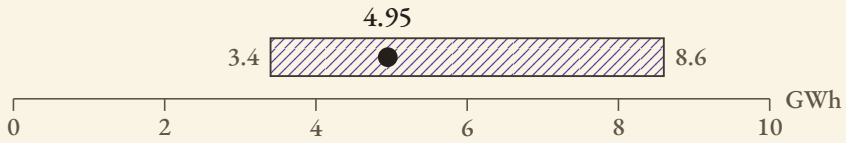
8.85 million GPU hours
90 days × 4,096 GPUs

×

560 W
per module

=

**4.95 GWh**




About 1,100 Swiss households for a year, at 4,500 kWh each.

*Range 750 to 1,900 households, over four sourced module figures: 560 W, tech report, arXiv 2509.14233; 660 W, CSCS cap, docs.cscs.ch/running/slurm; 855 W, from the TOP500 run; datasheet band 450 to 1,000 W. Household 4,500 kWh a year is ElCom's H4 profile. CSCS publish a PUE below 1.2, cooling from Lake Lugano (cscs.ch).*